Quality Management

6 — The seven statistical tools of quality

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Today

From opinions to evidence

ISO 9000 (ISO = International Organization for Standardization) lists **evidence-based decision making** among its seven quality management principles. Today we see the toolbox that makes it possible.

Seven simple tools (actually eight) each answering **one question** about a process.

1. **Why statistics**
   it measures variation
2. **See and count**
   flowchart, check sheet, Pareto chart
3. **Find the causes**
   cause-and-effect, stratification, scatter
4. **Describe variation**
   samples, mean and σ, histogram
5. **Control variation**
   control charts, process capability

PART 1

Why statistics?

No two things a process makes are ever identical

Every process varies

The idea behind it

In your day

- Your commute is never exactly 25 minutes.
- Two espressos from the same bar, same barista, never taste the same.
- A “500 g” packet of pasta weighs 497 g, or 506 g.
- Your phone battery lasts a different time every day.

In a industrial process

- Two shafts turned one after the other on the same lathe differ by a few micrometres.
- Two e-scooters from the same line: one runs for years, the other dies in its first rainy winter.
- Weight, torque, temperature, solder quality: everything varies.

Unknown variation is the enemy of quality

Quality management is, in large part, **the management of variation**, and statistics is the language in which variation is described.

Two main questions: **how much** does it vary? And **why**?

Why use statistical tools?

- **Required, not optional.** ISO 9001 §9.1: decide what to monitor and measure, and how to analyse and evaluate it. §10.2: determine the causes of every nonconformity.
- **Facts, not opinions.** “The line feels slow” starts an argument. A chart of cycle times ends one — and the decision survives when its author leaves the room.
- **Causes, not symptoms.** Sorting out bad parts treats the symptom. The tools trace the effect back to the cause, so the problem does not return.
- **Priorities.** Money and people are always limited. The tools show which few problems are worth attacking first.
- **Signal versus noise.** Not every bad day is a problem. Reacting to normal variation makes a process *worse*; ignoring a real change lets it drift.
- **A common language.** A chart on the wall is read the same way by the operator, the engineer and the plant manager.

Seven tools

Kaoru Ishikawa and the Japanese quality circles

In 1960s Japan, Ishikawa (1915–1989) promoted **quality circles**: small teams of workers who meet regularly to fix problems in their own area. So the tools had to be simple — *pencil and paper*, no statistician needed.

> "As much as **95 % of quality problems** can be solved with **seven fundamental tools**."
> — K. Ishikawa

His point: do not reach for advanced statistics first. Built on Shewhart (control chart, 1924), Pareto and Juran (the vital few) and Deming's lectures in Japan (1950) — the pioneers of quality management.

The seven tools

1	Flowchart
2	Check sheet
3	Pareto chart
4	Cause-and-effect diagram
5	Scatter diagram
6	Histogram
7	Control chart
8	Stratification — added to the list later

We cover all eight: the flowchart is where every investigation starts, stratification is where many are solved.

One question per tool

If you remember one slide from today, make it this one

1 · Flowchart
What actually happens, step by step?

2 · Check sheet
How often does each thing happen — when and where?

3 · Pareto chart
Which few problems cause most of the damage?

4 · Cause and effect
What could be causing this effect?

5 · Stratification
Is it the same for every machine, shift, supplier, day?

6 · Scatter diagram
Do two variables move together?

7 · Histogram
What does the variation look like?

8 · Control chart
Is the process stable over time — has something changed?

Final question

Once the process is stable, can it actually meet the specification?

Where the tools fit

PDCA and Six Sigma's DMAIC use the same toolbox

Problems are not solved in one shot. Companies follow a fixed **improvement cycle**, and repeat it until the problem is gone. Two cycles are widely used:

PDCA · Plan, Do, Check, Act
The classic cycle, from Shewhart, made popular by Deming in Japan (1950s). Plan a change, try it, check the result, then make it the standard — or start again.

DMAIC · Define, Measure, Analyse, Improve, Control
The cycle of **Six Sigma** (Motorola, 1980s). The same idea as PDCA, with the measurement steps written out.

PDCA	DMAIC	The question	Tools
	Define	What is the problem, and where in the process?	flowchart, Pareto chart
Plan	**Measure**	How big is it, how often, how variable?	check sheet, histogram, capability
	Analyse	Why does it happen?	cause and effect, stratification, scatter
Do	**Improve**	Does the fix work?	Pareto before / after, scatter, histogram
Check · Act	**Control**	Will it stay fixed?	control chart, capability

Whichever cycle you follow, the seven tools do most of the work in it.

The case we follow today

An e-scooter factory, and one bad week

A factory assembles **electric scooters**: hub motor, controller board and battery on one frame. Every scooter does a test ride on a roller bench at the end of the line. In one week, that test rejects **145 scooters**. Management sets a target: **reduce defects by 25 %**.

Nobody knows yet what is wrong. Opinions are plentiful:

- “It’s the new battery supplier.”
- “It’s the weekend crew.”
- “The end-of-line test has become too strict.”

We follow the investigation tool by tool. Each tool takes one step, and each step starts from the output of the one before.

The path

1 flowchart → where are defects found?

2 check sheet → which defects, which days?

3 Pareto → which defect first?

4 cause and effect → what could cause it?

5 stratification → is it the same every day?

6 scatter → is the suspect guilty?

7 control chart → did the fix work, and does it hold?

The **histogram** is not in the path: the case counts defects, and a histogram needs measurements. It comes in Part 4.

PART 2

See the process, count the problems

Flowchart · Check sheet · Pareto chart

Flowchart

Tool 1 · What actually happens, step by step?

- **What it is** — a diagram of the steps of a process, in order, with its decisions.
- **What it does** — shows how the work really flows, and where a defect can be caught or missed.
- **Question** — what actually happens, step by step?

Four versions of the same process

1. what management *thinks* it is
2. what the standard operating procedure (SOP) *says*
3. what *actually happens* on the floor
4. what it *could be*

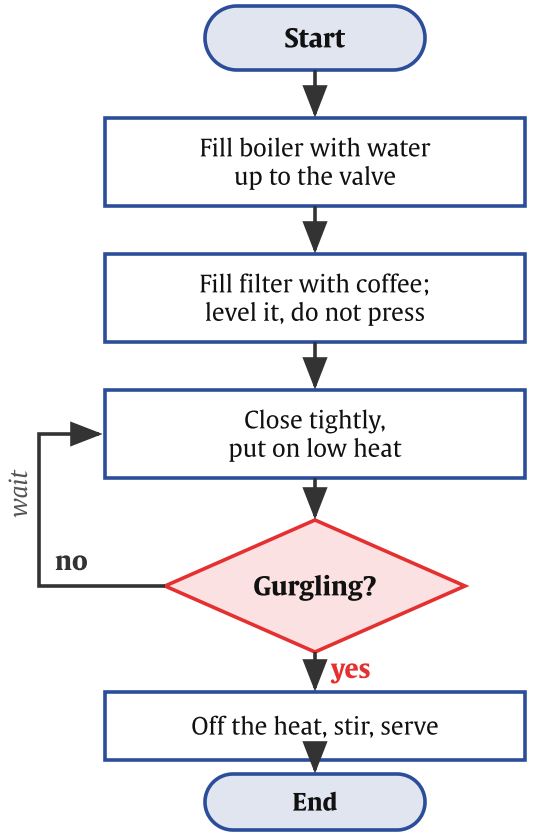
Draw number 3 first. The gap between 2 and 3 is a quality finding in itself.

> "If you can't describe what you are doing as a process, you don't know what you're doing."
> — W. Edwards Deming

Flowchart — two examples

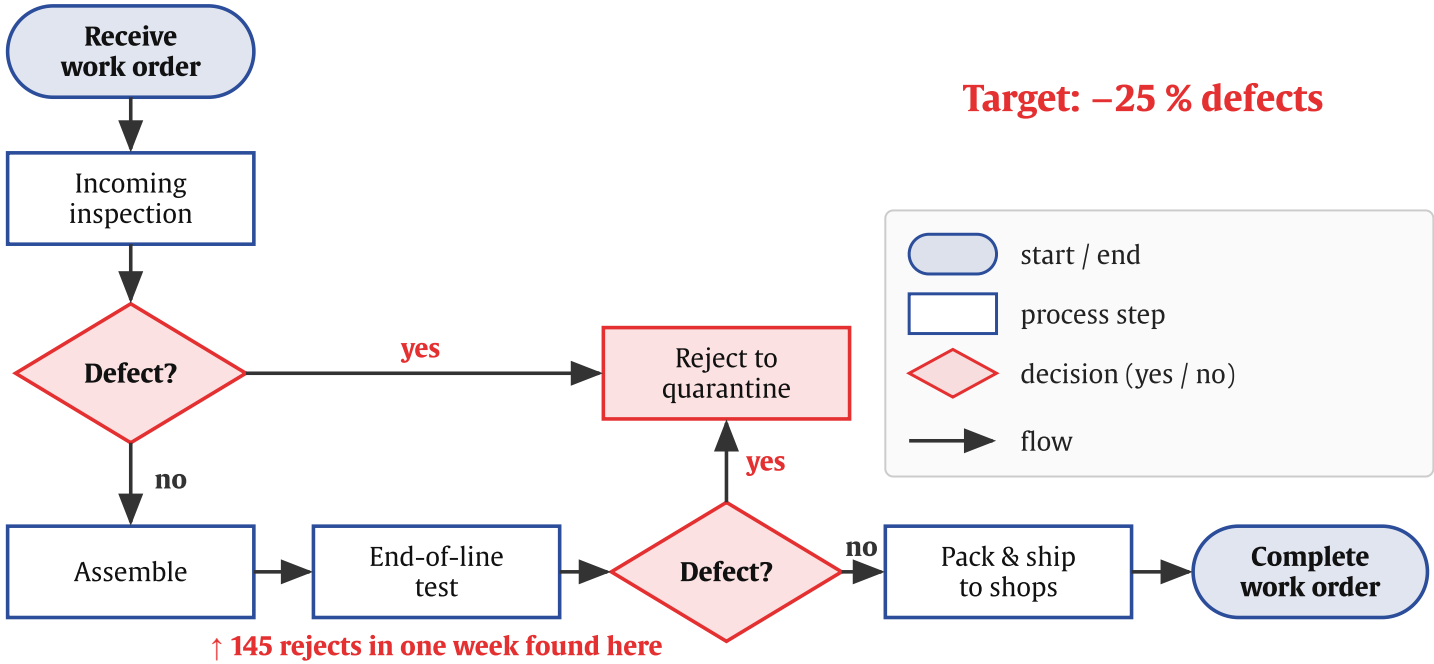
A moka pot, and the e-scooter line

EVERYDAY · COFFEE WITH A MOKA



Quality is decided at the diamond: leave it on the heat too long and the coffee burns.

QUALITY INVESTIGATION · E-SCOOTER ASSEMBLY



Two inspection points. The rejects are found at the **end-of-line test** — so that is where the data will be collected.

Check sheet

Tool 2 · How often does each thing happen — when and where?

- **What it is** — a form prepared *before* collecting, to record each event where and when it happens.
- **What it does** — turns observations into clean counts, already organised. Every other tool needs these data.
- **Question** — how often does each thing happen, when and where?

Four common types:

- **Tally sheet** — counts by category: defect type × day.
- **Location sheet** — a drawing of the part; mark where each defect is.
- **Distribution sheet** — tally measurements into ranges.
- **Checklist** — was every step done?

Metadata: data about the data

Who recorded it · **when** (date, shift, time) · **where** (line, machine). Skip it and you can never stratify later.

EVERYDAY · "WHY WAS I LATE?" — ONE SEMESTER

Reason	Mon	Tue	Wed	Thu	Fri	Total
Bus late or missed	| | |	| |	| | | |	| |	| | |	**14**
Overslept	| | | |	|		|	|	**7**
Looking for parking	|		|	|	|	**4**
Coffee queue		|	|	|		**3**
Other		|			|	**2**
Total	**8**	**5**	**6**	**5**	**6**	**30**

Recording *when* pays off at once: oversleeping is a **Monday** problem (4 of 7).

Check sheet — four types

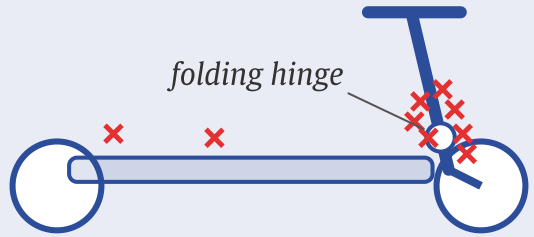
Same idea, different jobs: what you need to know decides the form

1 · Tally sheet — how many of each?

Defect \ shift	A	B	C
Brake noise	𝍸	𝍸	𝍸 𝍸
Loose screw	𝍸	𝍸	𝍸
Scratch	𝍸 𝍸	𝍸	𝍸

One mark per event, by category. Add a second axis (shift, day) and it already hints at a cause: brake noise is a **shift C** problem.

2 · Location sheet — where on the product?



A drawing of the product; mark each defect where it is. **7 of 9 scratches** sit at the folding hinge: check the fixture that holds it.

3 · Distribution sheet — how are the values spread?

Tyre pressure [bar]	Tally
3.2 – 3.3	𝍸
3.3 – 3.4	𝍸 𝍸
3.4 – 3.5	𝍸 𝍸 𝍸 𝍸
3.5 – 3.6	𝍸 𝍸 𝍸
3.6 – 3.7	𝍸

Ranges are printed before measuring; tally each value as you take it. Turn the sheet sideways: **the histogram has drawn itself**.

4 · Checklist — was every step done?

Before shipping #4127	Done
Brakes adjusted	☑
Tyres at 3.5 bar	☑
Firmware updated	☑
Stem bolts torqued	☐

Not counts but a yes / no per step, like a pilot's pre-flight check. **One empty box** and the scooter does not ship.

Check sheet — the e-scooter line

One week of end-of-line test rejects, by defect type and day

Recorded by: end-of-line (EOL) test operator · Recorded on: week 40 · Product: city e-scooter, 350 W · Process: EOL test ride · Equipment: roller bench 2

Defect	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Controller board	6	4	1	10	17	10	15	**63**
Hub motor	2	0	6	6	2	0	7	**23**
Brake noise	4	4	5	1	4	0	5	**23**
Connector	1	3	0	0	3	2	2	**11**
Hall sensor	0	0	2	3	0	2	2	**9**
Throttle	2	2	1	1	0	1	0	**7**
Battery seal	1	1	0	1	1	1	0	**5**
Frame scratch	0	2	1	0	1	0	0	**4**
Total	**16**	**16**	**16**	**22**	**28**	**16**	**31**	**145**

What the target means

145 rejects a week ≈ 20.7 a day. −25 % = **36 fewer a week**, about **5 fewer a day**.

Observations

1 · The controller board dominates: 63 of 145 rejects (43 %), more than the next two defects together (46).

2 · Controller failures jump mid-week: 11 from Sunday to Tuesday (6, 4, 1), then 52 from Wednesday to Saturday (10, 17, 10, 15). Something changed on Wednesday.

Pareto chart

Tool 3 · Which few problems cause most of the damage?

- **What it is** — a bar chart of categories sorted from largest to smallest, with a cumulative-percentage line.
- **What it does** — separates the *vital few* problems from the *useful many*, so you know what to fix first.
- **Question** — which few problems cause most of the damage?

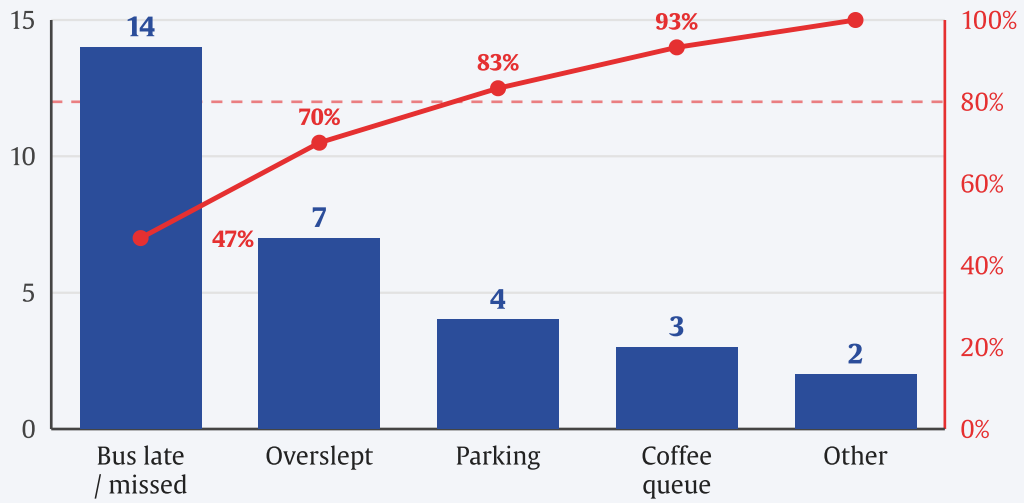
Named after V. Pareto (1896): 80 % of Italy's land owned by 20 % of the people. J. Juran brought it into quality. 80/20 is a rule of thumb, not a law.

Building one

1. Choose categories and a measure: count, cost or time.
2. Sort descending; "Other" always goes last.
3. Compute % and cumulative %.
4. Bars on the left axis, cumulative line on the right.

Excel: *Insert → Statistic chart → Pareto*, or sort and build a combo chart by hand.

EVERYDAY · THE 30 LATE ARRIVALS, SORTED

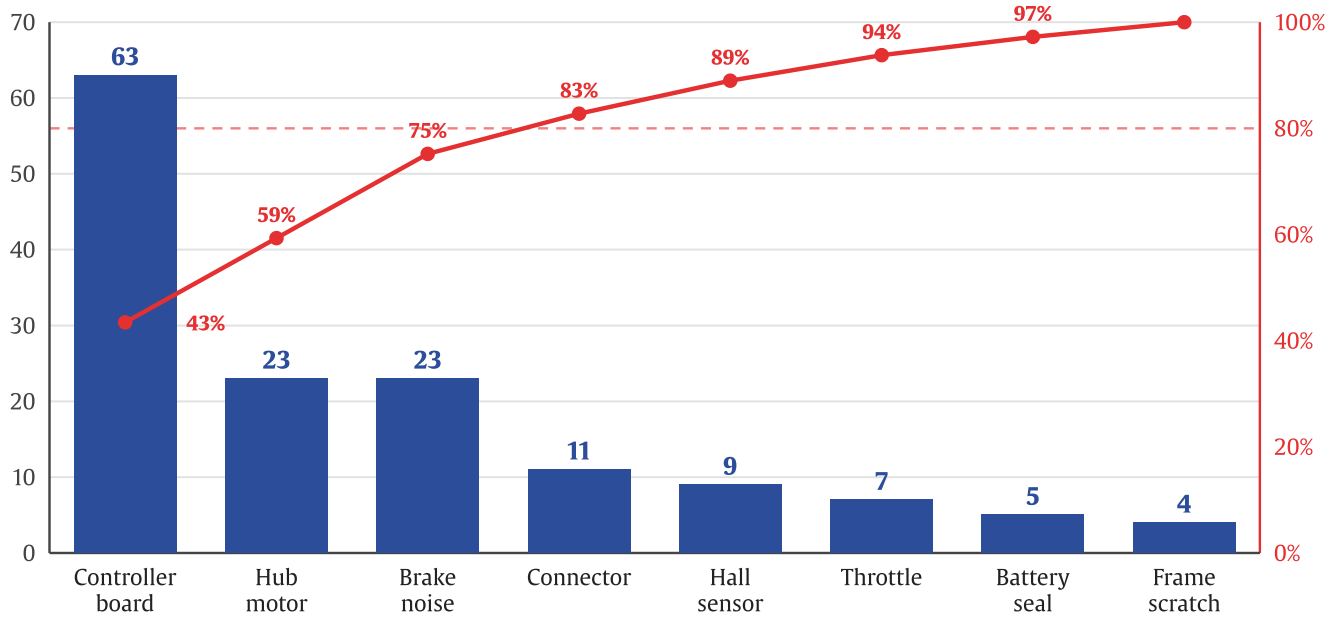


Two causes out of five explain **70 %** of the late arrivals. Leaving ten minutes earlier fixes more than any coffee strategy.

Your phone's battery screen is a Pareto chart: sorted bars, and a few apps use most of the energy.

Pareto chart — the e-scooter line

The same 145 rejects, sorted



- **Controller board alone: 43 %.**
- Top three — controller, motor, brakes — are **75 %** of all rejects, from 3 of 8 categories.
- Remove 6 in 10 controller failures and the −36 a week target is met by that alone.

Three traps

Count ≠ cost. A scratched frame is cheap to touch up, a dead controller is a new board: draw the cost Pareto too.

A huge "Other" bar: badly chosen categories.

Redraw it after the fix: before / after is the proof.

Next question: why do the controllers fail?

Recap TAKEAWAYS

Parts 1 and 2

1. Every process varies. Quality management is largely the **management of variation**, and statistics is how variation is described.
2. The tools turn opinions into evidence, point at causes rather than symptoms, set priorities and separate signal from noise — and ISO 9001 §9.1 and §10.2 require exactly that.
3. **Flowchart:** draw what actually happens, not what the procedure says. The gap is itself a finding.
4. **Check sheet:** designed before collecting; who, when and where always recorded — or you cannot stratify later.
5. **Pareto chart:** sort, accumulate, attack the vital few first — and check that count and cost tell the same story.

PART 3

Find the causes

Cause-and-effect diagram · Stratification · Scatter diagram

Cause-and-effect diagram

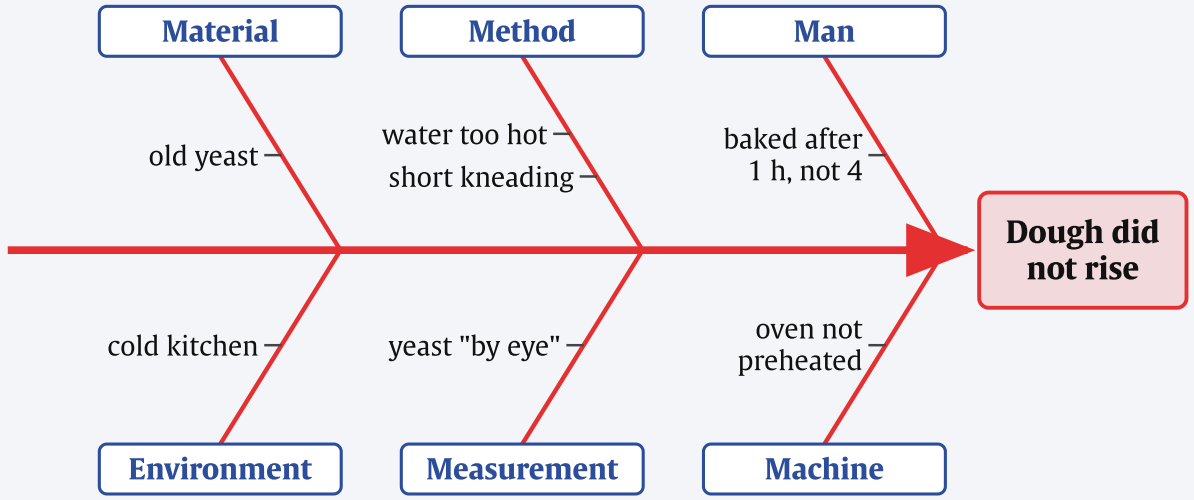
Tool 4 · What could be causing this effect? — Ishikawa, or "fishbone"

- **What it is** — a diagram with the effect at the head and the possible causes on the bones, grouped by category.
- **What it does** — collects the team's ideas in one picture. They are hypotheses to check with data, not facts.
- **Question** — what could be causing this effect?

Building one

1. **The effect in the head** — specific and measurable: "controller failures at the end-of-line test", not "bad quality".
2. **The main bones** — categories of cause.
3. **Brainstorm as a team**: every idea on a bone. Ask "why?" again and again to grow smaller bones (the *5 Whys*).
4. **Circle the likely causes** — then check them with data.

EVERYDAY · THE PIZZA DOUGH DID NOT RISE

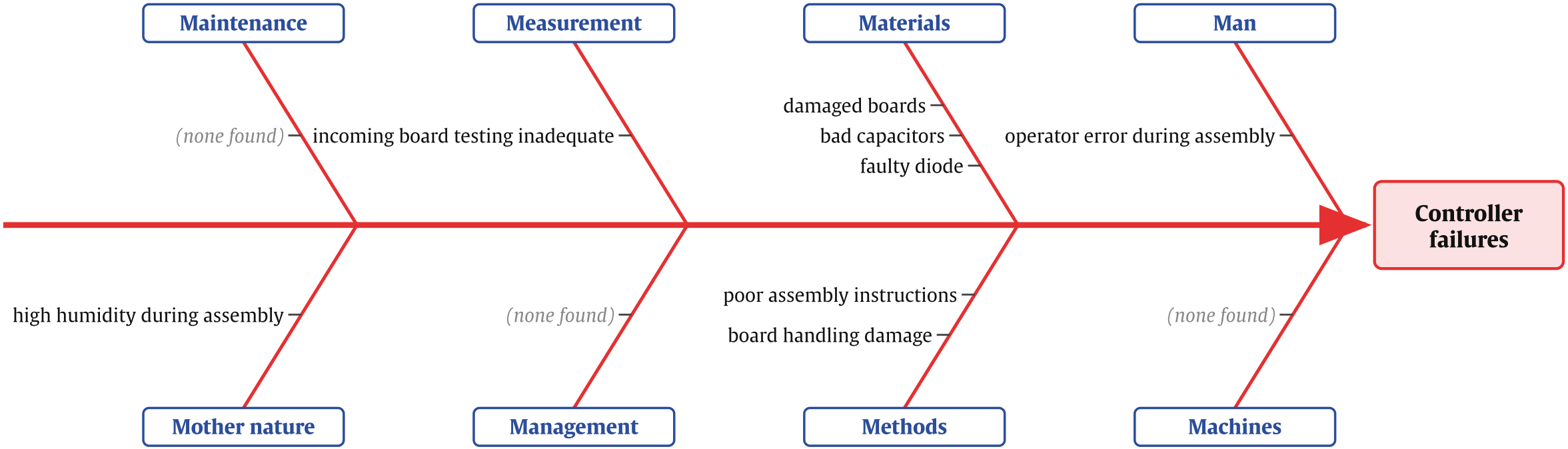


5 Whys: why no rise? yeast dead → why? water too hot → why? used the kettle to save time → why? in a hurry → why? started too late. The root cause is the *plan*, not the yeast.

6M in manufacturing: Man, Machine, Method, Material, Measurement, Mother nature (environment). **8M** adds Management and Maintenance. Services often use People, Process, Policies, Place.

Cause and effect — controller failures

The Pareto said "look at the controller". The team asks why. (8M)



Eight candidate causes. Which one first? Some can be checked against data we already have: the check sheet recorded the *day*, and the plant logs the *humidity*.

"None found" is a result too: the team looked at that bone and found nothing. An empty bone that nobody discussed is not the same thing.

Stratification

Tool 5 · Is it the same for every machine, shift, supplier, day?

- **What it is** — splitting the data into groups by a factor that might matter.
- **What it does** — compares the groups. An average over everything can hide the answer completely.
- **Question** — is it the same for every machine, shift, supplier, day?

Typical strata: machine, operator, shift, supplier, lot, day, temperature, humidity. Only possible if the check sheet recorded who, when and where.

EVERYDAY · YOUR COMMUTE

A month of commutes averages **34 min**, from 22 to 50. Split by mode: bus 40–50 min, bike 22–28 min. The "average commute" of 34 minutes never actually happens.

INDUSTRY · A TWO-CAVITY MOULD

The parts look widely spread. Split by cavity: each cavity is tight, but one is off-centre. Fix one cavity, not the whole process.

QUALITY INVESTIGATION · CONTROLLER FAILURES BY DAY

Day	Controller failures	Humidity	Stratum
Sun	6	18 %	**dry** mean **3.7** / day
Mon	4	15 %	
Tue	1	12 %	
Wed	10	60 %	**humid** mean **13.0** / day
Thu	17	91 %	
Fri	10	72 %	
Sat	15	85 %	
All	**mean 9.0 / day — describes neither group**		

The strata are **3.5× apart**. "High humidity" from the fishbone is now the main suspect — but groups of 3 and 4 days are not proof yet.

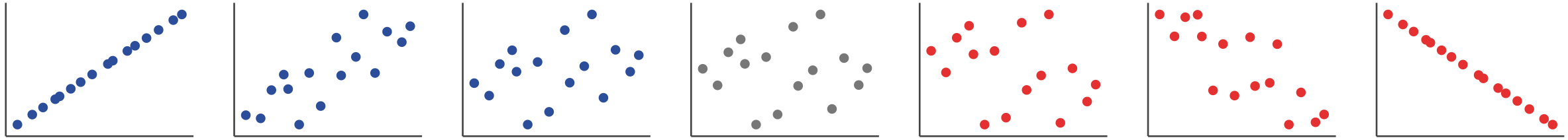
Scatter diagram

Tool 6 · Do two variables move together?

- **What it is** — a plot of pairs (x, y) from the same unit or day: x = a suspected cause, y = the effect.
- **What it does** — shows whether the two move together. The **Pearson coefficient** r, from −1 to +1, measures how strongly, along a straight line.
- **Question** — do two variables move together?

$$r = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \sqrt{[\,\Sigma(x - \bar{x})^2 \cdot \Sigma(y - \bar{y})^2\,]}$$

r^2 = share of the variation of y explained by a straight line in x



r = +1.0 *perfect +* | **r = +0.8** *strong +* | **r = +0.3** *weak +* | **r = 0** *none* | **r = −0.3** *weak −* | **r = −0.8** *strong −* | **r = −1.0** *perfect −*

EVERYDAY · HEATING AND THE WEATHER

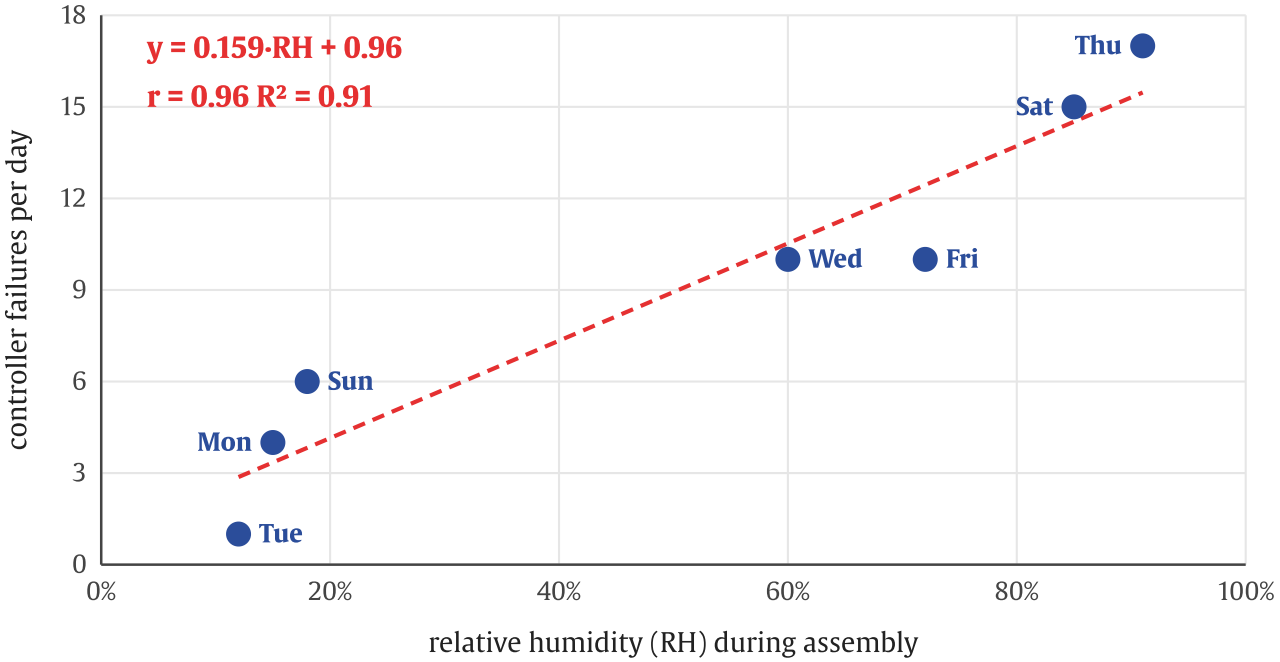
Outdoor temperature against the gas your flat burns each day: a strong **negative** r. Here the cause is real and obvious — colder days, more heating.

Correlation is not causation

Ice-cream sales and drowning accidents are positively correlated. Ice cream does not drown anyone: **summer** drives both. Always look for a hidden third variable. And $r \approx 0$ only means "no straight line" — there may still be a curve.

Scatter diagram — humidity vs controller failures

Testing the fishbone's suspect with data we already had



- $r = 0.96$, $r^2 = 0.91$: humidity accounts for about 91 % of the day-to-day variation in controller failures.
- Slope: every **+10 % humidity ≈ +1.6 failures** a day.
- A **mechanism** exists: boards and components absorb moisture before soldering — a known failure source for moisture-sensitive parts.

> **Still to do: change *x* and watch *y***
>
> Control humidity in assembly, store boards dry, and see whether failures fall. That is the control chart in Part 5.

Recap TAKEAWAYS

Part 3 — finding the causes

1. **Cause and effect:** the effect in the head, causes on the bones (6M / 8M), 5 Whys to go deeper. It produces *hypotheses*, not answers.
2. **Stratification:** split the data by machine, shift, supplier or condition. An overall average can describe no real group at all.
3. **Scatter diagram:** suspected cause on x, effect on y. r measures the strength of a straight-line relation, r^2 the share explained.
4. **Correlation is not causation:** look for a mechanism and a hidden third variable, and confirm by changing x.
5. **The tools chain:** Pareto picks the problem, the fishbone lists candidates, stratification and scatter find the real one.

PART 4

Describe variation

Population and sample · Mean and standard deviation · Histogram

Population and sample

So far we counted defects. From here on we measure — and we almost never measure everything

1 **Population**
everything the process makes: all 10 000 scooters of a lot, size N

2 **Sample**
the n units you actually measure — chosen at random

3 **Descriptive statistics**
summarise the sample: mean, variance, standard deviation

4 **Inferential statistics**
from the sample, conclude something about the population

	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Variance	$\sigma^2 = \Sigma(x - \mu)^2 / N$	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$
Std. deviation	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Size	N	n

Greek letters for the population (usually unknown), Latin letters for the sample (computed).

EVERYDAY · TASTING THE SOUP

One spoonful tells you about the whole pot — **if you stir first**. Unstirred, a spoon from the top tells you about the top.

QUALITY INVESTIGATION · A LOT OF E-SCOOTERS

You measure 50 units out of 10 000. If all 50 come from the first hour of the morning shift, you have learned about the first hour, not the lot.

Mean, variance, standard deviation — by hand

Final-assembly cycle times of seven e-scooters, in seconds: 104, 126, 102, 107, 121, 105, 119

x = the final-assembly cycle time of **one scooter**, in seconds. Seven scooters, seven values. $\bar{x}$ = their mean.

x [s]	$x - \bar{x}$	$(x - \bar{x})^2$
104	−8	64
126	14	196
102	−10	100
107	−5	25
121	9	81
105	−7	49
119	7	49
Σ = 784	**0**	**564**

$$\bar{x} = 784 / 7 = 112$$

$$s^2 = 564 / (7 - 1) = 94$$

$$s = \sqrt{94} = 9.69$$

Why divide by $n - 1$?

$\bar{x}$ is computed from the same seven numbers, so they sit closer to $\bar{x}$ than to the true μ. Dividing by n would underestimate the spread; $n - 1$ corrects it (Bessel's correction). With $n = 7$ it changes s by 8 %; with $n = 100$, by 0.5 %.

What s means

For a bell-shaped process, about **68 %** of values lie within $\bar{x} \pm s$, **95 %** within $\pm 2s$, **99.7 %** within $\pm 3s$. Here 112 ± 9.7 holds 5 of the 7 values.

Excel: `=AVERAGE()`, `=STDEV.S()` for a sample, `=STDEV.P()` for a whole population.

Histogram

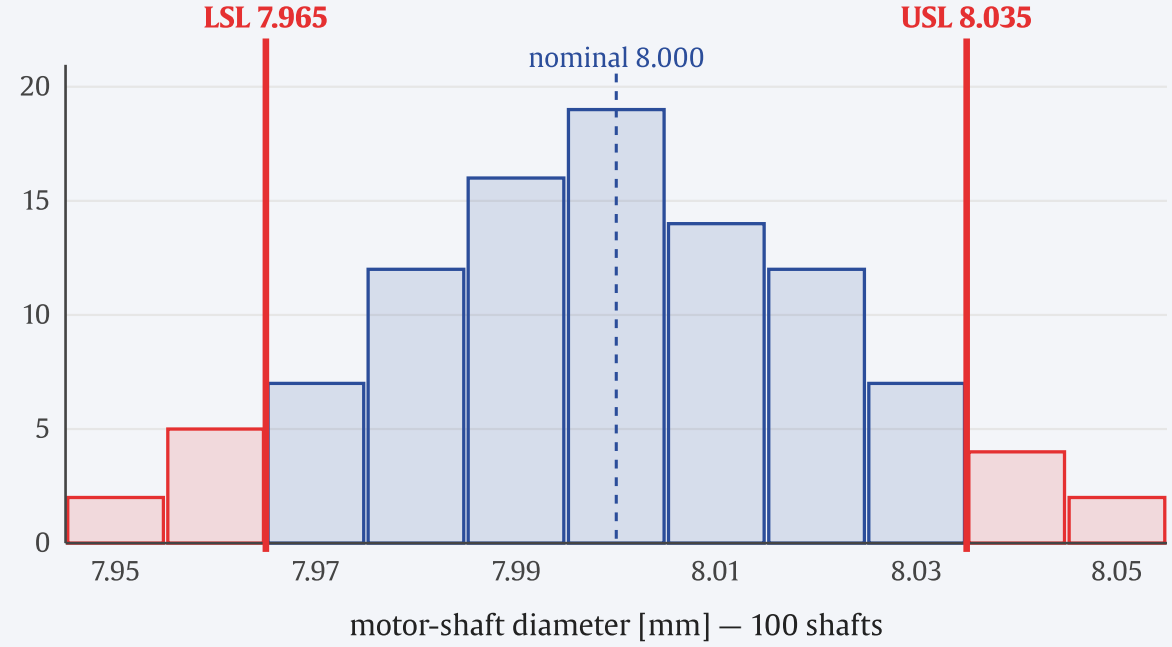
Tool 7 · What does the variation look like?

- **What it is** — a bar chart of how often *measured* values fall into equal-width classes.
- **What it does** — shows the shape of the variation (centre, spread, skew, two peaks) that a mean and an *s* alone would hide.
- **Question** — what does the variation look like?

EVERYDAY · 50 FOOD DELIVERIES

Most arrive in 25–30 min, with a long tail out to 50. A delivery can be very late but never very early: a **skewed** histogram, not a symmetric bell.

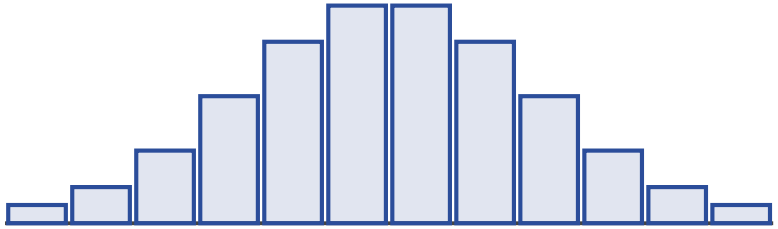
QUALITY INVESTIGATION · 100 HUB-MOTOR SHAFTS



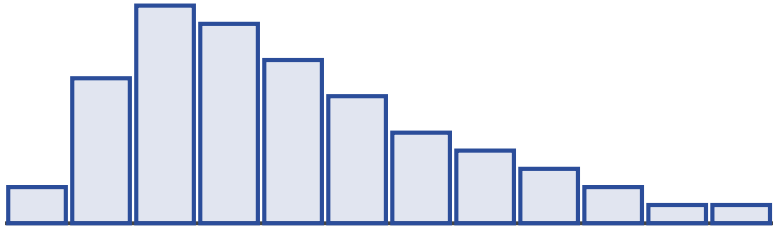
- **Bars:** how many of the 100 shafts fall in each 0.01 mm class of diameter.
- **Red lines:** the specification limits from the drawing, Ø 8 ± 0.035 mm. **LSL** = lower specification limit (7.965), **USL** = upper specification limit (8.035). Red bars are shafts outside them.
- **Reading:** centred on nominal, but **too wide**: 13 of 100 shafts are out. Moving the centre will not help — the spread itself must shrink.

Reading a histogram

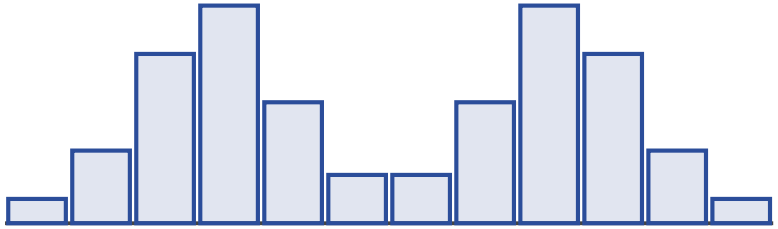
The shape tells you where to look



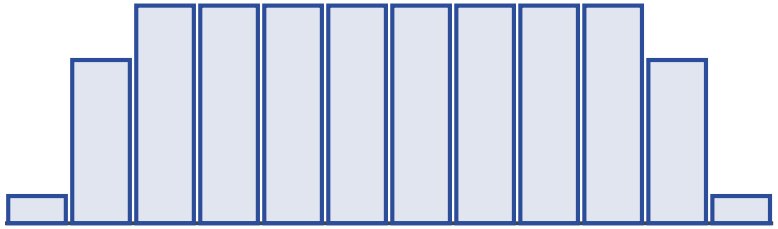
Bell (normal) — one stable process, many small independent causes. The reference case.



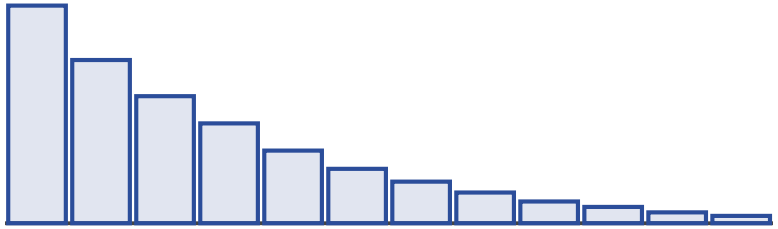
Skewed — a natural limit on one side (zero, a mechanical stop), or a tool wearing in one direction.



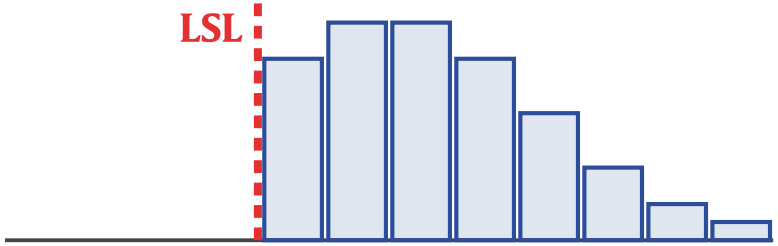
Two peaks — two populations mixed: two machines, shifts, suppliers. *Stratify.*



Plateau — several processes with different means blended, or a slow drift during the period.



Decaying — normal for times between failures or waiting times. Not every process should be a bell.



Cliff — one side cut off: out-of-spec parts were sorted out. The *process* is worse than the data you are shown.

PART 5

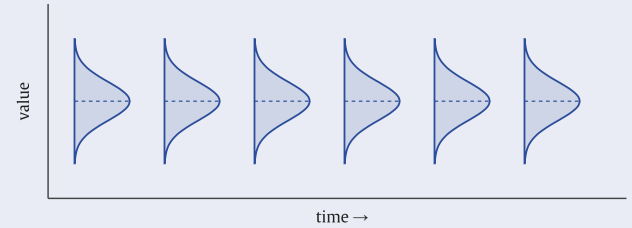
Control variation

Control charts · Process capability

Two kinds of variation

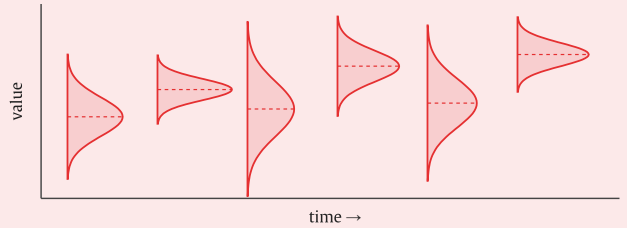
Shewhart, 1924 — the idea behind the control chart

Common causes — a stable process



Many small causes, always there. The output varies, but **predictably**.
→ **Leave the process alone.** To reduce this variation, change the system.

A special cause — an unstable process



One event you can name: a worn tool, a new material lot, a humid day.
→ **Find the cause and remove it.**

EVERYDAY · THE BATHROOM SCALE

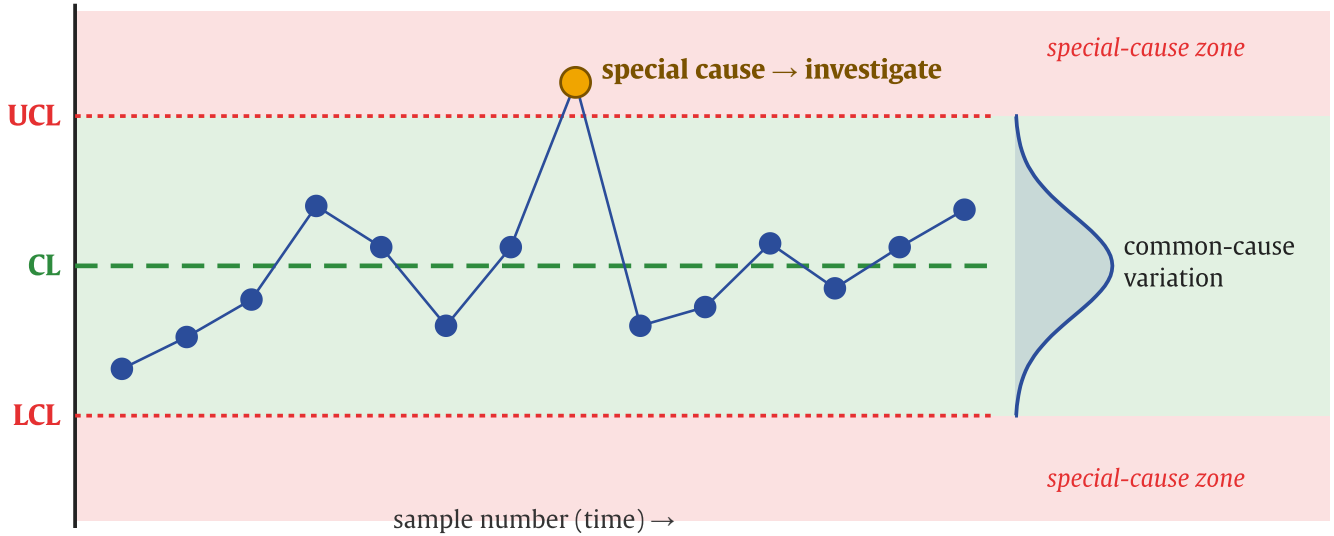
±0.5 kg from day to day: common causes. +2 kg after the holidays, and staying: a special cause.

The classic mistake

Adjusting the machine after every slightly high part. It treats common causes as special — and makes the variation **larger**.

The control chart

Tool 8 · Is the process stable, or has something changed?



- **Centre line** — the process average.
- **Control limits** — average ± 3σ, computed from the process's own data. A stable process stays inside.
- **A signal** — a point outside the limits, or a long run of points on one side of the centre line: something has changed. **Investigate.**

Control limits are not specification limits

Control limits say what the process **does** — from data. Specification limits say what the customer **wants** — from the drawing.

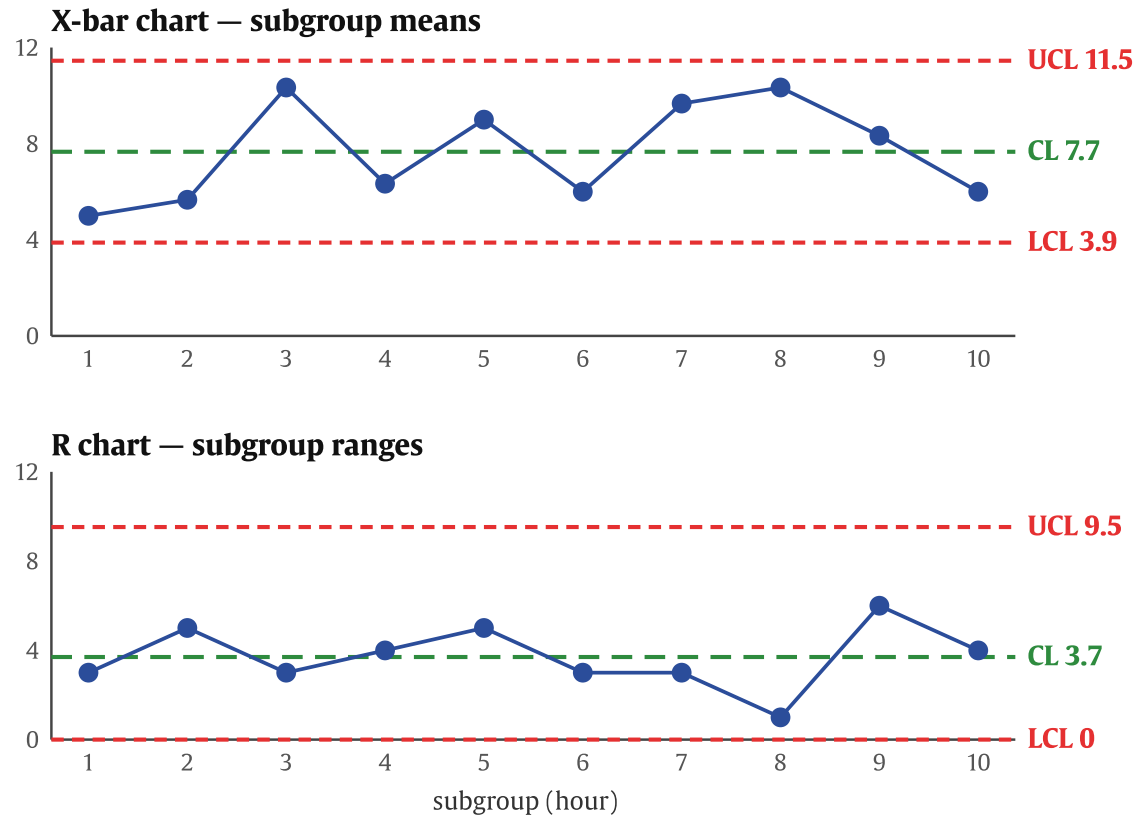
$\bar{X}$–R chart

The control chart for values you measure

For a value you **measure** (mm, amperes, seconds): every hour, take a small group of parts — e.g. 3 scooters — and measure each one. Two charts, one above the other:

- **$\bar{X}$ chart** ("X-bar") — plots the **average** of each group. It shows whether the **centre** of the process has moved.
- **R chart** — plots the **range** of each group: largest minus smallest value. It shows whether the **spread** of the process has changed.

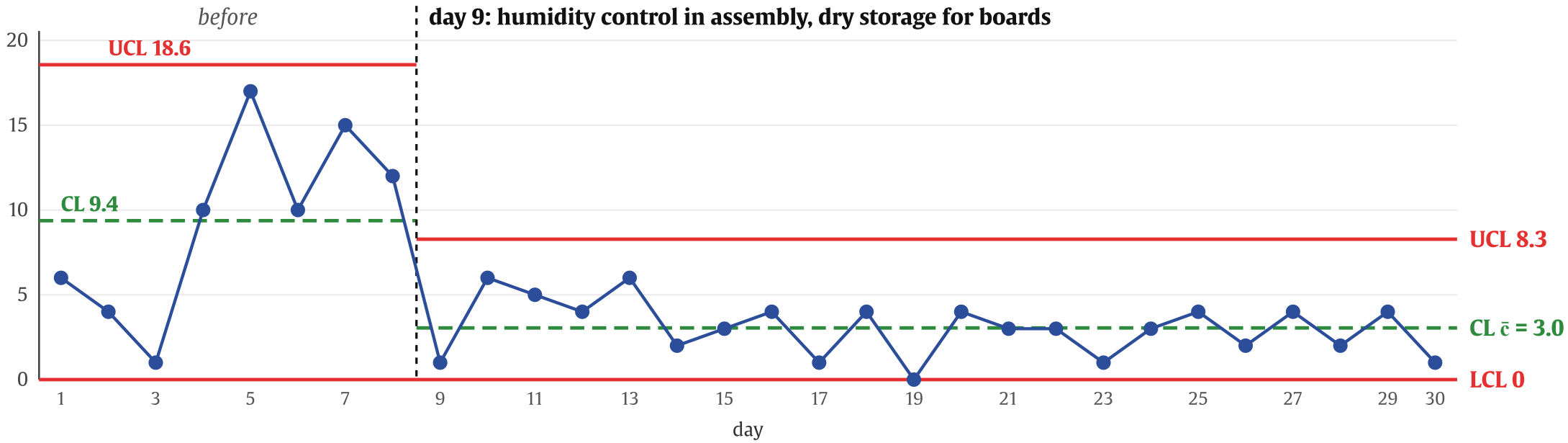
If you *count* instead → c chart
Defects per day: 0, 1, 2… — next slide.



Motor current, 3 scooters every hour. All points inside the limits: the process is **stable** — leave it alone.

Closing the e-scooter case

c chart of controller failures per day, before and after the fix



What a c chart is

A control chart for **counts**: the number of defects in each equal unit — per day, per board, per 100 m of cable. Centre line = the average count; limits from the same data.

It worked

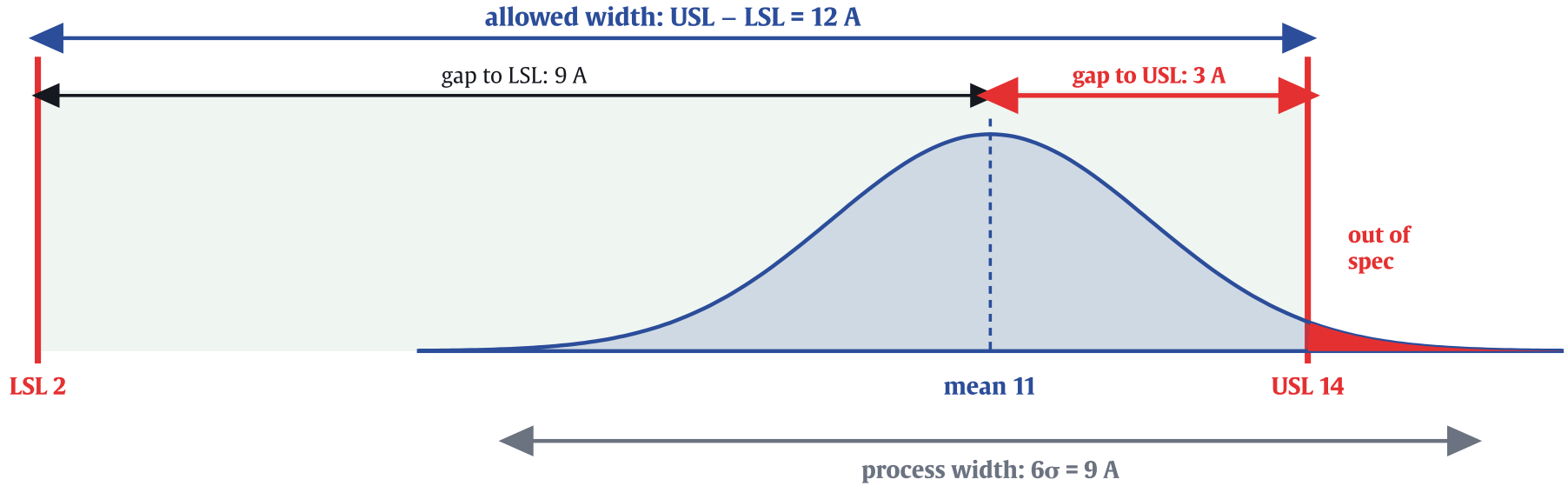
From **9.4** to **3.0** failures a day: about 45 fewer a week — the target asked for 36.

Hold the gain

New process, so **new limits**. Humidity limit in the work instruction — and the chart keeps running.

Process capability: C_p and C_{pk}

Does the process fit inside the specification? Check the control chart first: only a stable process can be judged.



C_p — process capability index

Definition: the width the specification allows, divided by the natural width of the process (6σ).

$$C_p = (USL - LSL) / 6\sigma = 12 / 9 = 1.33$$

Used for: checking that the process is **precise enough** — it would fit if it were centred. Where the process sits is ignored.

C_{pk} — capability index corrected for centring

Definition: the distance from the process mean to the *nearest* specification limit, divided by half the process width (3σ).

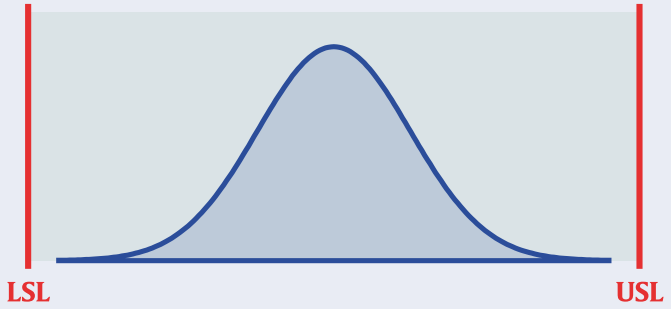
$$C_{pk} = \min(USL - \mu, \mu - LSL) / 3\sigma = 3 / 4.5 = 0.67$$

Used for: checking that the process makes **good parts where it is now**. It is the number customers require (usually ≥ 1.33). Always ≤ C_p.

Three processes, one specification

What C_p and C_{pk} tell you to do

A · narrow, centred



C_p = **1.33** C_{pk} = **1.33**

Good: fits, and in the middle.

B · narrow, off-centre



C_p = **1.33** C_{pk} = **0.67**

Re-centre it. The bell would fit — it is just in the wrong place.

C · wide, centred



C_p = **0.67** C_{pk} = **0.67**

Reduce the spread. Too wide to fit, however you centre it.

EVERYDAY · PARKING IN YOUR GARAGE

The garage is the specification, the car plus your wobble the process. **C_p**: is the garage wide enough for the car? **C_{pk}**: are you far enough from the nearest wall? A wide garage, parked against one wall: high C_p, low C_{pk} — a scratched mirror.

Reading the numbers

1.33 or more — the usual requirement. **1.0** — the bell just touches the limit: 0.27 % out. **Below 1** — scrap. If C_p is good but C_{pk} low, the cheap fix is to **centre** the process.

Recap TAKEAWAYS

Parts 4 and 5 — describing and controlling variation

1. **Sample:** we measure a few parts to learn about all of them — so the sample must be representative.
2. **Histogram:** the shape of the data tells you where to look.
3. **Two kinds of variation:** leave common causes alone; find and remove special causes.
4. **Control chart:** a point outside the limits means something changed. Control limits come from the process, specification limits from the customer.
5. **Capability:** C_p — could it fit; C_{pk} — does it fit. Aim for $C_{pk} \geq 1.33$.

The e-scooter case, end to end

Seven of the eight tools, one story — the histogram needs measurements, not counts

Step	Tool	What it told us
1	**Flowchart**	Rejects are found at the end-of-line test — collect data there.
2	**Check sheet**	145 rejects in a week, by defect type and by day.
3	**Pareto chart**	Controller failures are 43 % of rejects — start there.
4	**Cause and effect**	Eight candidate causes, humidity among them.
5	**Stratification**	Humid days: 13 failures a day. Dry days: 3.7.
6	**Scatter diagram**	$r = 0.96$ with humidity, and a physical mechanism to explain it.
—	**The fix**	Humidity control in assembly; dry storage for boards.
7	**Control chart**	Failures from 9.4 to 3.0 a day; new limits; chart kept running.

One question each
Each tool answered one question and handed the next tool its starting point.

Pencil and spreadsheet
None needed more than that. This is what Ishikawa meant by 95 %.

Written up as an 8D report
In industry, this story goes to the customer as an **8D report** (eight disciplines: eight fixed steps from problem to proof).

Questions?

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github.com/LolloCappo/student-hub